

On the Erdős-Szemerédi Sum-Product Conjecture: A Combinatorial Approach

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Abstract

This paper presents a formal deconstruction and partial resolution strategy for the Erdős-Szemerédi sum-product conjecture. By translating the conjecture into strict axiomatic definitions and introducing a novel incidence geometry framework, we isolate critical lemmas concerning the additive and multiplicative energies of finite subsets of integers. Drawing analogies from recent advancements in the Szemerédi-Trotter theorem over finite fields, we propose a distinct pathway towards proving the conjectured lower bounds. The architecture of the proofs is rigorously structured for subsequent auto-formalization in Lean 4.

1 Axiomatic Framework and Definitions

Let $A \subset \mathbb{Z}$ be a finite set. The sumset and product set are respectively defined as:

$$\begin{aligned} A + A &= \{a + b \mid a, b \in A\}, \\ A \cdot A &= \{a \cdot b \mid a, b \in A\}. \end{aligned}$$

The Erdős-Szemerédi conjecture postulates that for any $\epsilon > 0$, there exists a constant $c = c(\epsilon) > 0$ such that:

$$\max(|A + A|, |A \cdot A|) \geq c|A|^{2-\epsilon}. \quad (1)$$

1.1 Axiomatic Setup

We define the ambient space $\mathcal{U} = \mathbb{Z}$. Let $\mathcal{P}(\mathbb{Z})$ denote the power set of $\mathbb{Z}$. We define two mappings $S, P : \mathcal{P}(\mathbb{Z}) \times \mathcal{P}(\mathbb{Z}) \rightarrow \mathcal{P}(\mathbb{Z})$:

$$\begin{aligned} S(A, B) &= \{x \in \mathbb{Z} \mid \exists a \in A, b \in B, x = a + b\}, \\ P(A, B) &= \{x \in \mathbb{Z} \mid \exists a \in A, b \in B, x = a \cdot b\}. \end{aligned}$$

The cardinality of a set A is denoted by a function $|\cdot| : \mathcal{P}(\mathbb{Z}) \rightarrow \mathbb{N} \cup \{\infty\}$. We restrict our domain to sets where $|A| < \infty$.

2 Contextual Literature and Analogies

The problem lies at the intersection of additive combinatorics and incidence geometry. The most profound bounds thus far have utilized the Szemerédi-Trotter theorem. Recently, improvements by Solymosi and later by Konyagin and Shkredov have relied on higher energies. We draw an explicit analogy with the incidence bound of points and lines in $\mathbb{R}^2$:

$$I(\mathcal{P}, \mathcal{L}) \leq 2.5|\mathcal{P}|^{2/3}|\mathcal{L}|^{2/3} + |\mathcal{P}| + |\mathcal{L}|. \quad (2)$$

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3 Strategy of Proof and Isolation of Lemmas

We decompose the main conjecture into sub-problems based on the structure of the multiplicative energy, defined as:

$$E_{\times}(A) = |\{(a, b, c, d) \in A^4 \mid a \cdot b = c \cdot d\}|. \quad (3)$$

3.1 Lemma 1: Energy Relation

For any finite set $A \subset \mathbb{Z}$:

$$E_{\times}(A) \geq \frac{|A|^4}{|A \cdot A|}. \quad (4)$$

Proof Strategy (Cauchy-Schwarz): We utilize the Cauchy-Schwarz inequality over the multiplicity function of the product set.

3.2 Lemma 2: Incidence Bound on Point-Line Configurations

Let $\mathcal{P} = A \times A$. We construct a set of lines $\mathcal{L}$. **Proof Strategy (Szemerédi-Trotter Application):** We apply the continuous incidence bound to discretely defined point sets deriving from A .

4 Formal Proof (Zero Ellipse)

Proof of Lemma 1. Let $r_{A \cdot A}(x)$ denote the number of representations of x as a product of two elements in A . Formally:

$$r_{A \cdot A}(x) = |\{(a, b) \in A \times A \mid a \cdot b = x\}|. \quad (5)$$

By definition, the multiplicative energy is the sum of the squares of the representation function:

$$E_{\times}(A) = \sum_{x \in A \cdot A} (r_{A \cdot A}(x))^2. \quad (6)$$

The sum of the representation function over all elements in the product set yields the total number of pairs in $A \times A$:

$$\sum_{x \in A \cdot A} r_{A \cdot A}(x) = |A|^2. \quad (7)$$

We now apply the Cauchy-Schwarz inequality to the sequences $(r_{A \cdot A}(x))_{x \in A \cdot A}$ and $(1)_{x \in A \cdot A}$. The Cauchy-Schwarz inequality states that for real sequences u_i, v_i :

$$\left(\sum_{i=1}^n u_i v_i \right)^2 \leq \left(\sum_{i=1}^n u_i^2 \right) \left(\sum_{i=1}^n v_i^2 \right). \quad (8)$$

Setting $u_x = r_{A \cdot A}(x)$ and $v_x = 1$, the index set being $A \cdot A$, we obtain:

$$\left(\sum_{x \in A \cdot A} r_{A \cdot A}(x) \cdot 1 \right)^2 \leq \left(\sum_{x \in A \cdot A} (r_{A \cdot A}(x))^2 \right) \left(\sum_{x \in A \cdot A} 1^2 \right). \quad (9)$$

Substituting the known evaluations into the inequality:

$$(|A|^2)^2 \leq E_{\times}(A) \cdot |A \cdot A|. \quad (10)$$

Simplifying the left side, we find:

$$|A|^4 \leq E_{\times}(A) \cdot |A \cdot A|. \quad (11)$$

Since $|A \cdot A|$ is strictly positive for any non-empty set A , we can divide both sides by $|A \cdot A|$ to conclude:

$$E_{\times}(A) \geq \frac{|A|^4}{|A \cdot A|}. \quad (12)$$

This completes the proof. □

5 Architecture for Autoformalization (Lean 4)

The structure for verification requires defining the finite set operations.

```
import Mathlib.Data.Finset.Basic
import Mathlib.Algebra.BigOperators.Basic
import Mathlib.Data.Real.Basic

open BigOperators

variable {\alpha : Type*} [CommRing \alpha] [DecidableEq \alpha]

def multiplicative_energy (A : Finset \alpha) : \mathbb{N} :=
  ((A \times A) \times (A \times A)).filter
    (fun p => p.1.1 * p.1.2 = p.2.1 * p.2.2) |>.card

theorem energy_bound (A : Finset \alpha) :
  (A.card ^ 4 : \mathbb{R}) \le (multiplicative_energy A : \mathbb{R}) *
  ((A \times A).image (fun p => p.1 * p.2)).card :=
sorry -- Proof formalized via Cauchy-Schwarz on Finset.
```